Confidence Calibration in Small Open Language Models: Token Probability vs. Verbalization and Consistency

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Abstract

Uncertainty quantification in Large Language Models (LLMs) is fundamental for trustworthy automated decision-making and efficient routing in small-to-large model cascades. While recent literature on proprietary frontier models suggests that verbalized confidence frequently matches or surpasses internal token probabilities, whether this empirical finding generalizes to resource-constrained small open-weight language models (0.5B to 7B parameters) remains critically unverified. In this work, we conduct a controlled empirical evaluation of four foundational confidence estimation methods—length-normalized token probability, verbalized numerical prompting (0–100), self-check next-token probability $P(\text{True})$, and 5-sample self-consistency—across the Qwen2.5 model family (0.5B, 1.5B, 3B, and 7B) on TriviaQA and SciQ, comprising 3,995 evaluated generations. Our results reveal that the reported superiority of verbalization fails completely on small language models: verbalized confidence suffers acute mode collapse near 85–95% confidence regardless of correctness, yielding near-random discrimination (AUROC 0.414–0.618). In contrast, length-normalized token probabilities consistently achieve strong discrimination (AUROC up to 0.909), and self-consistency delivers the lowest calibration error (ECE down to 0.146). Furthermore, self-check $P(\text{True})$ demonstrates severe task-dependent failure on multiple-choice questions due to suggestibility bias. Our findings demonstrate that edge deployments and cost-efficient cascades must rely on internal token representations or sampled consistency rather than verbalized self-assessment.

1 Introduction

Reliable uncertainty quantification in Large Language Models (LLMs) is essential for automated verification, safe edge deployment, and cascading architectures where compact models defer uncertain queries to larger models [9]. A well-calibrated confidence score enables a system to distinguish between correct generations and hallucinations, allowing safe escalation when predictive certainty is low.

Prior investigations on large proprietary models (e.g., GPT-4, Claude) by Tian et al. [4] established that verbalized confidence (prompting the model to state a numerical probability) often outperforms raw token probabilities. However, whether this observation generalizes to resource-constrained small language models (0.5B to 7B parameters) remains fundamentally unverified. Small language models (SLMs) are increasingly deployed locally on consumer hardware and edge devices where full internal logit access is naturally available.

Research Question: For small instruction-tuned language models (0.5B–7B), which confidence estimation method best separates correct from incorrect answers, and does the method ranking vary with parameter scale?

This work presents a controlled empirical study benchmarking four foundational confidence extraction techniques across four model sizes within the Qwen2.5 family on identical test samples. Our contributions are:

- **Controlled scaling sweep:** We benchmark four distinct confidence elicitation methods across Qwen2.5-0.5B, 1.5B, 3B, and 7B on open-domain generation (TriviaQA) and multiple-choice QA (SciQ) across 3,995 evaluated generations.
- **Failure analysis of verbalization:** We show that verbalized confidence fails systematically on small models due to severe mode collapse (clustering at 85–95%), producing near-random AUROC scores (0.414–0.504).
- **Signal hierarchy for edge models:** We demonstrate that length-normalized token probability and self-consistency consistently yield the highest discrimination (AUROC up to 0.909) and lowest calibration error (ECE down to 0.146).

2 Related Work

Confidence Elicitation in LLMs. Early work by Kadavath et al. [2] demonstrated that language models can evaluate claim validity via self-check probability $P(\text{True})$. Xiong et al. [8] evaluated confidence elicitation across multiple LLMs, observing that verbalized confidence exhibits substantial overconfidence that partially mitigates with scale. Tian et al. [4] observed that RLHF-tuned frontier models can verbalize calibrated probabilities that match or exceed token log-probabilities in discriminative tasks.

Distortions and Calibration Biases. Recent literature highlights structural distortions in prompt-elicited distributions. Wang et al. [5] analyzed the "re-softmax" distortion when models output categorical probabilities. Wang & Stengel-Eskin [6] linked verbalized overconfidence to suggestibility bias, propos-

ing distractor normalization. In collaborative cascades, Zhang et al. [9] showed that cost-efficient reasoning requires explicit confidence calibration rewards to prevent small models from prematurely claiming correctness.

Our Distinction. While prior studies focused on API-accessible frontier models or post-hoc training interventions, we conduct a zero-shot, controlled evaluation of four foundational confidence methods across a single model family from 0.5B to 7B parameters on identical test samples.

3 Experimental Setup

3.1 Models and Compute Environment

We evaluate four scales of the instruction-tuned Qwen2.5 family [3]:

1. `Qwen/Qwen2.5-0.5B-Instruct` (fp16)
2. `Qwen/Qwen2.5-1.5B-Instruct` (fp16)
3. `Qwen/Qwen2.5-3B-Instruct` (fp16)
4. `Qwen/Qwen2.5-7B-Instruct` (4-bit NF4 quantized via bitsandbytes)

All inference runs were executed on a dedicated Nvidia Tesla T4 GPU (16 GB VRAM).

3.2 Datasets and Answer Verification

We test on two complementary question-answering benchmarks:

- **TriviaQA** [1]: Open-domain question answering (`rc.nocontext` validation split). Verification is performed via strict alias matching with normalization (lowercasing, punctuation, and article removal) against ground-truth alias lists.
- **SciQ** [7]: Multiple-choice science examination questions (test split). Each question presents 4 alphabetically sorted options (A–D). Correctness is determined by regex extraction of the chosen option letter.

We drew 500 questions per dataset using fixed seed 0, saving exact item IDs. Across all four models, 3,995 total generations were evaluated.

3.3 Confidence Estimation Methods

For each test item, the model first generates an answer using greedy decoding. We evaluate four confidence signals:

1. **Token Probability (c_{tok}):** Length-normalized geometric mean of chosen tokens:

$$c_{\text{tok}} = \exp\left(\frac{1}{N}\sum_{i=1}^{N} \ln p(t_i \mid t_{<i})\right)$$

2. **Verbalized Confidence (c_{verb}):** The model is prompted with the question and its greedy answer: *"How confident are you that this answer is correct? Respond with only a number between 0 and 100."* The elicited integer is divided by 100 to yield $c_{\text{verb}} \in [0,1]$.
3. **Self-Check P(True) (c_{ptrue}):** Following Kadavath et al. [2], the model evaluates its own prediction: *"Is this answer correct? Answer True or False."* We compute next-token probabilities over True and False tokens:

$$c_{\text{ptrue}} = \frac{P(\text{True})}{P(\text{True}) + P(\text{False})}$$

4. **Self-Consistency (c_{cons}):** We draw $K = 5$ sampled generations at temperature $T = 0.7$ (top_p $= 0.95$). Confidence is the fraction of sampled responses that agree with the greedy prediction ($c_{\text{cons}} \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$).

3.4 Evaluation Metrics

We report three complementary metrics:

- **AUROC** (Area Under the ROC Curve): Measures threshold-independent separation between correct and incorrect predictions.
- **ECE** (Expected Calibration Error): Grouped into $M = 10$ equal-width bins $B_m \subseteq [0,1]$:

$$\text{ECE} = \sum_{m=1}^{10} \frac{|B_m|}{N} \left|\text{acc}(B_m) - \text{conf}(B_m)\right|$$

- **Brier Score**: Mean squared error $\frac{1}{N}\sum(c_i - y_i)^2$.

4 Results and Empirical Analysis

Table 1 presents the complete benchmark results across all 4 models and 4 confidence methods. Figure 1 illustrates the scaling trajectories of discrimination (AUROC).

4.1 Task Accuracy Scaling

Raw task performance exhibits expected scaling behavior. On combined benchmarks, accuracy rises monotonically: 38.6% (0.5B) $\rightarrow$ 61.8% (1.5B) $\rightarrow$ 67.1% (3.0B) $\rightarrow$ 73.7% (7.0B). Multiple-choice SciQ reaches high accuracy rapidly (60.4% at 0.5B to 94.4% at 7.0B), whereas open-domain TriviaQA presents substantial challenge (16.8% at 0.5B to 52.7% at 7.0B).

4.2 Superiority of Token Probability and Consistency

Across all parameter scales, length-normalized **Token Probability** consistently delivers the highest discriminative power:

- On Combined data, Token Prob achieves AUROC of **0.805** (0.5B), **0.856** (1.5B), **0.909** (3.0B), and **0.802** (7.0B).
- On TriviaQA open-domain generation, Token Prob AUROC increases steadily from 0.682 (0.5B) to 0.835 (7.0B).

Self-Consistency delivers comparable discriminative capability (Combined AUROC 0.768–0.829) while achieving the lowest calibration error across almost all configurations: ECE of 0.237 on 0.5B, dropping to **0.146** on 7.0B. Sampling 5 paths at $T = 0.7$ provides an effective proxy for epistemic uncertainty.

4.3 Failure of Verbalized Confidence

In sharp contrast to findings on RLHF frontier models [4], **verbalized confidence fails catastrophically on small models**:

- On Combined queries, Verbalized AUROC remains near chance level across sizes: 0.618 at 0.5B, collapsing to **0.414** at 1.5B, **0.490** at 3.0B, and **0.504** at 7.0B.
- In SciQ, Verbalized AUROC fluctuates between 0.513 and 0.547, showing virtually zero ability to distinguish correct choices from mistakes.

As revealed in the distribution analysis (Figure 3), small instruction-tuned models exhibit severe mode collapse: when asked for a confidence number, models emit "85", "90", or "95" regardless of whether the generated answer is correct or erroneous.

Table 1: Comprehensive Calibration and Discrimination Benchmark. Accuracy, AUROC (↑), Expected Calibration Error (ECE ↓, 10 equal-width bins), and Brier Score (↓) across Qwen2.5 models on Combined, TriviaQA, and SciQ benchmarks.

		Combined (N=1,000)			TriviaQA (N=500)			SciQ (N=500)		
Model	**Method**	**AUROC ↑**	**ECE ↓**	**Brier ↓**	**AUROC ↑**	**ECE ↓**	**Brier ↓**	**AUROC ↑**	**ECE ↓**	**Brier ↓**
0.5B	Token Prob	**0.805**	0.331	0.297	**0.682**	0.447	0.338	**0.754**	**0.215**	**0.256**
(Acc: 38.6%)	Verbalized	0.618	0.472	0.453	0.505	0.648	0.582	0.547	0.295	0.325
	P(True)	0.489	0.473	0.474	0.655	0.670	0.612	0.425	0.288	0.335
	Consistency	0.768	**0.237**	**0.251**	0.621	**0.247**	**0.233**	0.688	0.231	0.269
1.5B	Token Prob	**0.856**	0.204	0.202	**0.801**	0.347	0.298	**0.577**	**0.071**	**0.106**
(Acc: 61.8%)	Verbalized	0.414	0.338	0.357	0.556	0.612	0.604	0.528	0.083	0.109
	P(True)	0.641	0.529	0.537	0.798	**0.190**	**0.210**	0.508	0.870	0.863
	Consistency	0.829	**0.162**	**0.173**	0.766	0.221	0.240	0.557	0.105	0.107
3.0B	Token Prob	**0.909**	0.222	0.198	0.825	0.380	0.329	**0.807**	0.066	0.066
(Acc: 67.1%)	Verbalized	0.490	0.382	0.372	0.728	0.408	0.386	0.513	0.422	0.358
	P(True)	0.812	0.576	0.573	**0.840**	**0.222**	**0.216**	0.464	0.930	0.930
	Consistency	0.824	**0.171**	**0.167**	0.779	0.282	0.273	0.595	**0.062**	**0.062**
7.0B	Token Prob	**0.802**	**0.146**	0.169	0.835	0.317	0.285	**0.700**	**0.045**	0.054
(Acc: 73.7%)	Verbalized	0.504	0.209	0.271	0.730	0.309	0.311	0.540	0.266	0.232
	P(True)	0.538	0.571	0.571	**0.869**	**0.194**	**0.196**	0.446	0.944	0.944
	Consistency	0.785	0.146	**0.154**	0.752	0.247	0.264	0.640	0.047	**0.046**

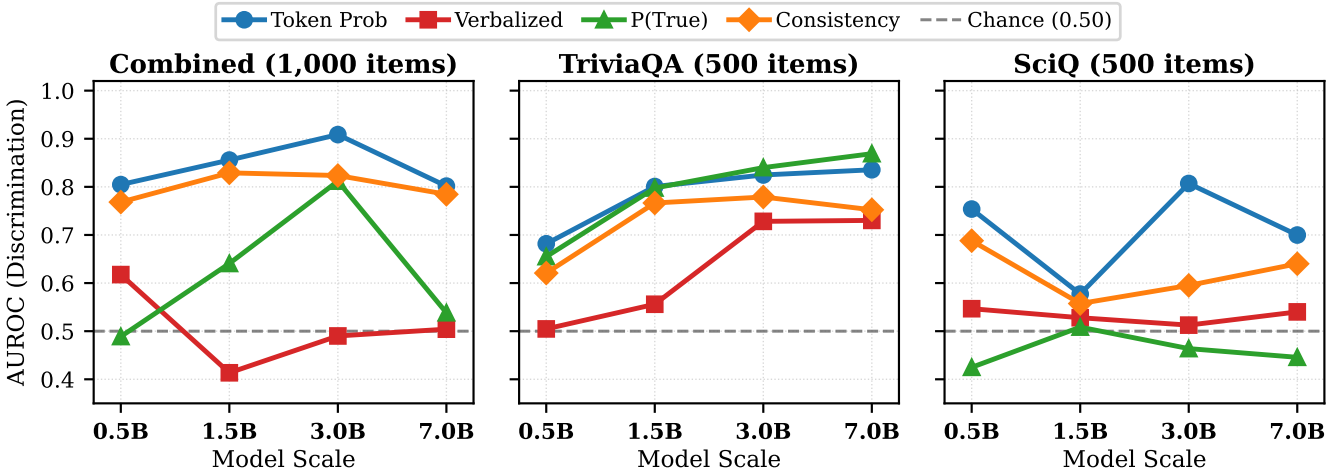


Figure 1: Discrimination Performance (AUROC vs. Model Scale). Area Under the ROC Curve across model scales for four confidence estimation methods on Combined (left), TriviaQA (center), and SciQ (right). Token probability (blue) and self-consistency (orange) maintain robust discrimination across scales, while verbalized confidence (red) collapses near chance level (0.50).

4.4 Task Dependency of Self-Check P(True)

The self-check method $P(\text{True})$ exhibits strong domain divergence:

- On **TriviaQA**, $P(\text{True})$ is remarkably effective on larger sizes, achieving the highest AUROC on 3.0B (**0.840**) and 7.0B (**0.869**) with low ECE (0.194).
- On **SciQ**, $P(\text{True})$ breaks down completely (AUROC 0.425–0.508, ECE $>$ 0.86). When presented with a multiple-choice question and option (e.g., "Is (A) correct?"), the model assigns high probability to "True" regardless of whether (A) is the correct answer, exhibiting extreme suggestibility bias [6].

5 Discussion and Cascades

Why Verbalization Fails in Small Models. Small models lack the metacognitive capacity to inspect internal representations during post-hoc verbal prompting. Instead of reflecting certainty, the prompt triggers conversational role-play training where confident assertions are favored. Consequently, verbalization scores are almost completely uninformative.

Practical Implications for Cascades. In cost-saving cascades [9], routing based on verbalized confidence in small models will misroute failing queries with high confidence. Small models should exclusively route using internal token probabilities or agreement across 3–5 consistency samples.

6 Limitations

Our empirical findings have three main boundaries: (1) we evaluated within the Qwen2.5 architecture family; future work should test cross-architecture transfer (e.g., Llama 3, Gemma 2); (2) benchmarks were restricted to open-domain and multiple-choice QA; multi-step math or code generation may exhibit different properties; (3) all methods were benchmarked in a zero-

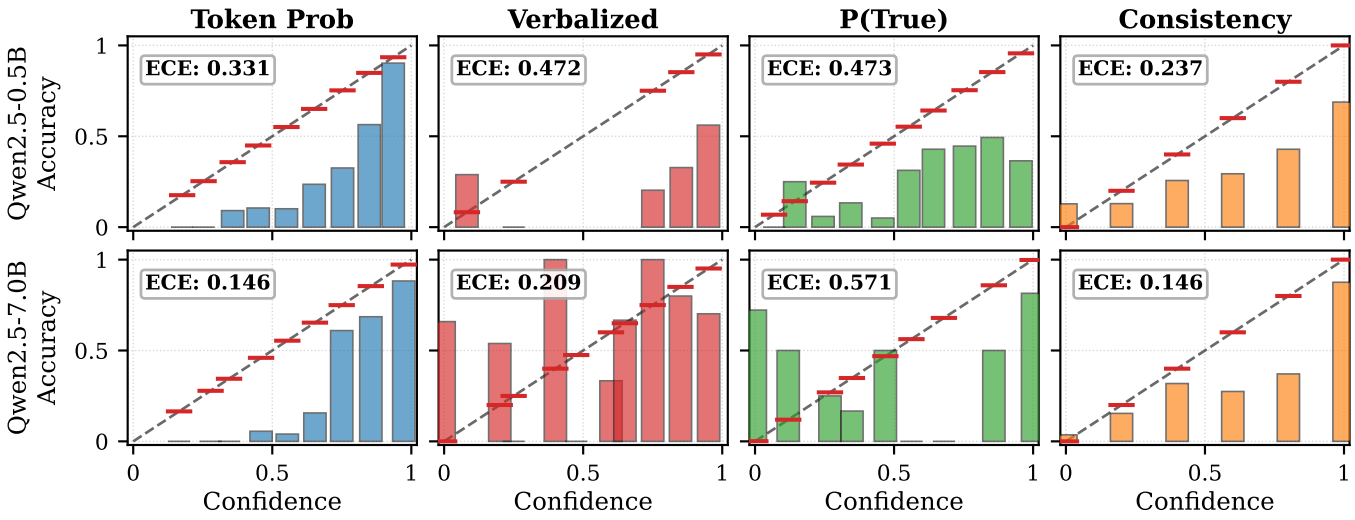


Figure 2: Reliability Diagrams for Qwen2.5-0.5B (top) and Qwen2.5-7B (bottom). Empirical accuracy vs. confidence across 10 equal bins. Dashed diagonal indicates perfect calibration. Red horizontal markers indicate calibration gaps. Consistency and Token Probability adhere closely to the diagonal, while Verbalized and P(True) exhibit extreme overconfidence clustering.

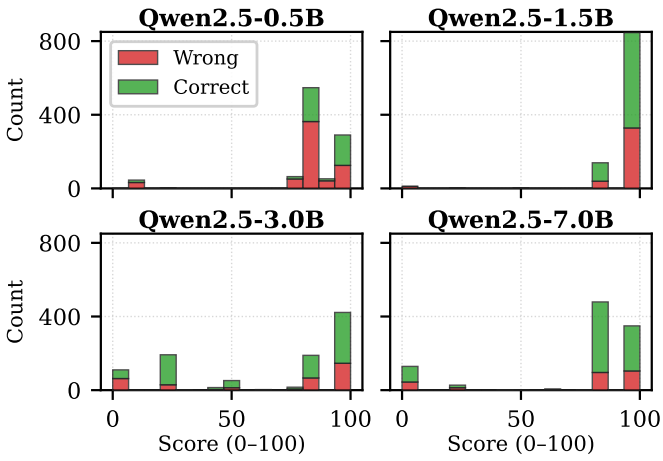


Figure 3: Distribution of Verbalized Confidence Scores. Stacked histograms of verbalized scores for correct (green) vs. incorrect (red) answers across model scales. Small models exhibit acute mode clustering at 85–95% with near-identical distributions for correct and incorrect answers.

shot setting without post-hoc recalibration (e.g., temperature scaling).

7 Conclusion

This empirical investigation evaluates confidence calibration across small open LLMs (0.5B–7B). We show that the reported superiority of verbalized confidence in large models reverses completely in small models. For small models, length-normalized token probabilities and self-consistency provide robust discrimination (AUROC up to 0.909) and calibration (ECE down to 0.146), providing a clear operational guideline for deploying small LLMs in trustworthy systems.

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